

A Guide to pip-run: Features, Installation, and Usage

pip-run provides on-demand temporary package installation for a single execution run.

It replaces this series of commands (or their Windows equivalent):

```
$ virtualenv --python pythonX.X --system-site-packages $temp/env  
$ $temp/env/bin/pip install pkg1 pkg2 -r reqs.txt  
$ $temp/env/bin/python ...  
$ rm -rf $temp/env
```

With this single-line command:

```
$ py -X.X -m pip-run pkg1 pkg2 -r reqs.txt -- ...
```

py is the Python Launcher for Windows/Unix. It's not required to use pip-run, but it's recommended because it makes running Python in a clean, portable way much easier.

Features include

- Automatically finds dependencies from imports.
- Downloads missing packages and makes them available for use.
- Installs packages temporarily (they are removed when the process ends).
- Uses pip's cache to avoid re-downloading packages.
- Leaves no permanent traces (except cached files).
- Can replace installed packages if needed.
- Re-uses the standard pip installation process.

What pip-run is for

pip-run is not meant for production dependency management. Instead, it's designed for quick, one-time tasks like:

- Running scripts with dependencies
- Testing or experimenting
- Build setup
- Test running
- Interactive development
- Bug fixing / triage

Installation

Install pip-run like a normal Python package:

```
pip install pip-run
```

It works in both system Python and virtual environments.

Usage

You can use pip-run in different ways:

- **Script launcher** → Run scripts with temporary dependencies.
- **Dependency context manager** → Add packages just for that run.
- **Interactive interpreter** → Open Python shell with chosen packages.
- **Module launcher** → Like `python -m`.
- **Shebang in scripts** → `#!/usr/bin/env pip-run` lets you make single-file tools.

It will help to reduce the overflow error to resolve this we can add some more information inside this box.

How to Run

Run from the command line:

```
pip-run
```

Or run as a module (handy for multiple Python versions):

```
python -m pip-run
```

Note: Run `pip-run --help` for full details.